Cholinergic Drugs

Learning Outcomes

List types of cholinergic agonists
Recognize pharmacologic effects of cholinergic agonists
Enumerate therapeutic uses of cholinergic agonists
Describe the manifestations and treatment of overdose of anticholinesterases

Parasympathetic (Cholinergic)Nervous System

NEUROTRANSMITTER (Neurohormone)

Acetylcholine – Produced and stored in nerve endings

Acetylcholine (ACh) binds to Cholinergic receptor sites and causes a response

Stimulation results in “rest and digest’

Acetylcholinesterase(AchE)
(cholinesterase)

Breaks down Acetylcholine or limits response

Maintains the balance of acetylcholine

Introduction to Cholinergic Drugs

Definition: Drugs that mimic or enhance the action of acetylcholine, a neurotransmitter.

Relevance in Dentistry: Impact on salivary secretion, muscle control, and autonomic regulation.

Classification:

- Direct-acting cholinergic agonists
- Indirect-acting cholinergic agents (Cholinesterase inhibitors)

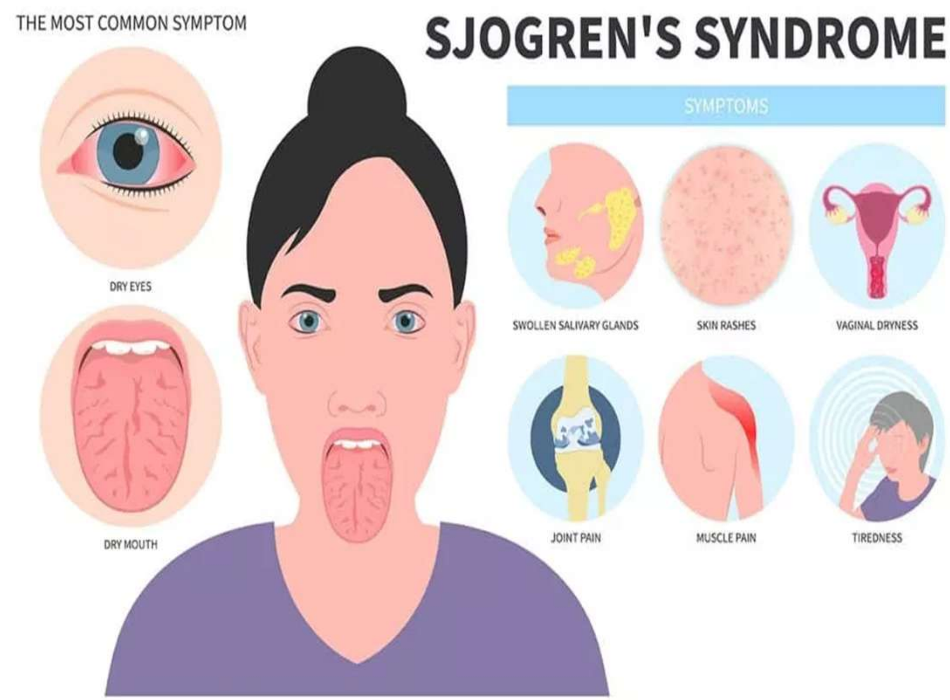
Direct-Acting Cholinergic Agonists

Examples:

- Acetylcholine
- Pilocarpine (used to treat dry mouth in dentistry)
- Bethanechol

Mechanism of Action: Mimic acetylcholine by directly stimulating cholinergic receptors.

Clinical Applications: Glaucoma treatment, Sjögren's syndrome in dental practice



Indirect-Acting Cholinergic Agents (Cholinesterase Inhibitors)

Examples:

- Neostigmine
- Physostigmine

Mechanism: Inhibit acetylcholinesterase, increasing acetylcholine levels.

Clinical Relevance in Dentistry: Managing muscle relaxation post-anesthesia, possible role in myasthenia gravis.

Clinical Uses of Cholinergic Drugs in Dentistry

Managing Xerostomia: Pilocarpine stimulates salivary glands in a dry mouth.

Anesthesia Recovery: Agents like neostigmine for muscle relaxation post-surgery.

Contraindications: Considerations in patients with asthma, peptic ulcer, and heart conditions.

Adverse Effects

Common Side Effects:

- Excess salivation
- Sweating
- Diarrhea
- Bradycardia

Dental Considerations: How to manage these effects in clinical settings.

Cholinergic Crisis

Definition:	Symptoms:	Emergency Management:
• Overstimulation of cholinergic receptors.	• Muscle weakness • Respiratory depression • Excessive secretion	• Use of anticholinergic drugs like atropine.

Anticholinergic Drugs in Dentistry

Contrast with Cholinergic Drugs: Reduce saliva, manage bradycardia, used during surgery.

Examples: Atropine, Scopolamine

Irreversible Inhibition of AchE

Some of the compounds irreversibly inhibits enzyme acetylcholinesterase (which breaks down acetylcholine), thereby increasing acetylcholine levels.

Examples: Organophosphate Insecticides (Malathion) & Organophosphate-containing nerve agents (Sarin).

Symptoms: Depression of respiration and circulation, tremor, general weakness, paralysis and potentially coma.

Treatment: Pralidoxime(Reactivates AchE enzyme) and Atropine(Blocks muscarinic receptors)

Books to Read

- Bart –Jhonson, Frank J. Dowd. Pharmacology and Therapeutic for Dentistry, 6th edition, 2011. Elsevier Publishers,USA
- Karen Whalen, Richard Finkel, Thomas A Panavelil. Lippincott Illustrated Reviews Pharmacology. 6th ed. 2015. Philadelphia Wolter Kluwer Puplisher.